

Contrastes paramétricos

Parámetro	Varianza conocida y otras condiciones	Valor observado del Estadístico de contraste	Contraste de hipótesis	Región de aceptación de H_0	p-valor
μ	Si	$Z_o = \frac{\bar{x} - \mu_o}{\sigma/\sqrt{n}}$	$H_0: \mu = \mu_o$ vs $H_A: \mu < \mu_o$	$(-z_{\alpha}, +\infty)$	$P\{Z \leq Z_o\}$
			$H_0: \mu = \mu_o$ vs $H_A: \mu > \mu_o$	$(-\infty, z_{\alpha})$	$P\{Z \geq Z_o\}$
			$H_0: \mu = \mu_o$ vs $H_A: \mu \neq \mu_o$	$(-z_{\alpha/2}, z_{\alpha/2})$	$P\{ Z \geq Z_o \} = 2P\{Z \geq Z_o \}$
	No $n \geq 30$	$Z_o = \frac{\bar{x} - \mu_o}{s/\sqrt{n}}$	$H_0: \mu = \mu_o$ vs $H_A: \mu < \mu_o$	$(-z_{\alpha}, +\infty)$	$P\{Z \leq Z_o\}$
			$H_0: \mu = \mu_o$ vs $H_A: \mu > \mu_o$	$(-\infty, z_{\alpha})$	$P\{Z \geq Z_o\}$
			$H_0: \mu = \mu_o$ vs $H_A: \mu \neq \mu_o$	$(-z_{\alpha/2}, z_{\alpha/2})$	$P\{ Z \geq Z_o \} = 2P\{Z \geq Z_o \}$
	No $n < 30$ Distribución Normal	$t_o = \frac{\bar{x} - \mu_o}{s/\sqrt{n}}$	$H_0: \mu = \mu_o$ vs $H_A: \mu < \mu_o$	$(-t_{\alpha, n-1}, +\infty)$	$P\{t \leq t_o\}$
			$H_0: \mu = \mu_o$ vs $H_A: \mu > \mu_o$	$(-\infty, t_{\alpha, n-1})$	$P\{t \geq t_o\}$
			$H_0: \mu = \mu_o$ vs $H_A: \mu \neq \mu_o$	$(-t_{\alpha/2, n-1}, t_{\alpha/2, n-1})$	$P\{ t \geq t_o \} = 2P\{t \geq t_o \}$
p	$n \geq 30$	$Z_o = \frac{\hat{p} - p_o}{\sqrt{\frac{p_o(1-p_o)}{n}}}$	$H_0: p = p_o$ vs $H_A: p < p_o$	$(-z_{\alpha}, +\infty)$	$P\{Z \leq Z_o\}$
			$H_0: p = p_o$ vs $H_A: p > p_o$	$(-\infty, z_{\alpha})$	$P\{Z \geq Z_o\}$
			$H_0: p = p_o$ vs $H_A: p \neq p_o$	$(-z_{\alpha/2}, z_{\alpha/2})$	$P\{ Z \geq Z_o \} = 2P\{Z \geq Z_o \}$

Siendo:

$$z_{\frac{\alpha}{2}} = \text{cuantil}\left(1 - \frac{\alpha}{2}\right) \text{ de } Z: N(0,1) \rightarrow P\left\{Z \leq z_{\frac{\alpha}{2}}\right\} = 1 - \frac{\alpha}{2} \quad (\text{Tabla A4})$$

$$z_{\alpha} = \text{cuantil}(1 - \alpha) \text{ de } Z: N(0,1) \rightarrow P\{Z \leq z_{\alpha}\} = 1 - \alpha \quad (\text{Tabla A4})$$

$$t_{\frac{\alpha}{2}, n-1} = \text{cuantil}\left(1 - \frac{\alpha}{2}\right) \text{ de } T: \text{Student}(n-1) \rightarrow P\left\{T \leq t_{\frac{\alpha}{2}, n-1}\right\} = 1 - \frac{\alpha}{2} \rightarrow P\left\{T > t_{\frac{\alpha}{2}, n-1}\right\} = \frac{\alpha}{2} \quad (\text{Tabla A5})$$

$$t_{\alpha, n-1} = \text{cuantil}(1 - \alpha) \text{ de } T: \text{Student}(n-1) \rightarrow P\{T \leq t_{\alpha, n-1}\} = 1 - \alpha \rightarrow P\{T > t_{\alpha, n-1}\} = \alpha \quad (\text{Tabla A5})$$

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

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Diferencia de parámetros	Varianzas conocidas y otras condiciones	Valor observado del Estadístico de contraste	Contraste de hipótesis	Región de aceptación de H_0	p-valor
$\mu_X - \mu_Y$ Muestras independientes	Si	$Z_o = \frac{\bar{x} - \bar{y} - D}{\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}}}$	$H_0: \mu_X - \mu_Y = D$ vs $H_A: \mu_X - \mu_Y < D$	$(-z_{\alpha}, +\infty)$	$P\{Z \leq Z_o\}$
			$H_0: \mu_X - \mu_Y = D$ vs $H_A: \mu_X - \mu_Y > D$	$(-\infty, z_{\alpha})$	$P\{Z \geq Z_o\}$
			$H_0: \mu_X - \mu_Y = D$ vs $H_A: \mu_X - \mu_Y \neq D$	$(-z_{\alpha/2}, z_{\alpha/2})$	$P\{ Z \geq Z_o \} = 2P\{Z \geq Z_o \}$
	No $n, m \geq 30$	$Z_o = \frac{\bar{x} - \bar{y} - D}{\sqrt{\frac{s_X^2}{n} + \frac{s_Y^2}{m}}}$	$H_0: \mu_X - \mu_Y = D$ vs $H_A: \mu_X - \mu_Y < D$	$(-z_{\alpha}, +\infty)$	$P\{Z \leq Z_o\}$
			$H_0: \mu_X - \mu_Y = D$ vs $H_A: \mu_X - \mu_Y > D$	$(-\infty, z_{\alpha})$	$P\{Z \geq Z_o\}$
			$H_0: \mu_X - \mu_Y = D$ vs $H_A: \mu_X - \mu_Y \neq D$	$(-z_{\alpha/2}, z_{\alpha/2})$	$P\{ Z \geq Z_o \} = 2P\{Z \geq Z_o \}$
	No $n, m < 30$ $\sigma_X \neq \sigma_Y$ Distribución Normal	$t_o = \frac{\bar{x} - \bar{y} - D}{\sqrt{\frac{s_X^2}{n} + \frac{s_Y^2}{m}}}$	$H_0: \mu_X - \mu_Y = D$ vs $H_A: \mu_X - \mu_Y < D$	$(-t_{\alpha, v}, +\infty)$	$P\{t \leq t_o\}$
			$H_0: \mu_X - \mu_Y = D$ vs $H_A: \mu_X - \mu_Y > D$	$(-\infty, t_{\alpha, v})$	$P\{t \geq t_o\}$
			$H_0: \mu_X - \mu_Y = D$ vs $H_A: \mu_X - \mu_Y \neq D$	$(-t_{\alpha/2, v}, t_{\alpha/2, v})$	$P\{ t \geq t_o \} = 2P\{t \geq t_o \}$
	No $n, m < 30$ $\sigma_X = \sigma_Y$ Distribución Normal	$t_o = \frac{\bar{x} - \bar{y} - D}{s_p \sqrt{\frac{1}{n} + \frac{1}{m}}}$	$H_0: \mu_X - \mu_Y = D$ vs $H_A: \mu_X - \mu_Y < D$	$(-t_{\alpha, n+m-2}, +\infty)$	$P\{t \leq t_o\}$
			$H_0: \mu_X - \mu_Y = D$ vs $H_A: \mu_X - \mu_Y > D$	$(-\infty, t_{\alpha, n+m-2})$	$P\{t \geq t_o\}$
			$H_0: \mu_X - \mu_Y = D$ vs $H_A: \mu_X - \mu_Y \neq D$	$(-t_{\alpha/2, n+m-2}, t_{\alpha/2, n+m-2})$	$P\{ t \geq t_o \} = 2P\{t \geq t_o \}$
$\mu_{(X-Y)}$ Muestras pareadas	No $n \geq 30$	$Z_o = \frac{(\overline{X-Y}) - D}{s_{(X-Y)}/\sqrt{n}}$	$H_0: \mu_{(X-Y)} = D$ vs $H_A: \mu_{(X-Y)} < D$	$(-z_{\alpha}, +\infty)$	$P\{Z \leq Z_o\}$
			$H_0: \mu_{(X-Y)} = D$ vs $H_A: \mu_{(X-Y)} > D$	$(-\infty, z_{\alpha})$	$P\{Z \geq Z_o\}$
			$H_0: \mu_{(X-Y)} = D$ vs $H_A: \mu_{(X-Y)} \neq D$	$(-z_{\alpha/2}, z_{\alpha/2})$	$P\{ Z \geq Z_o \} = 2P\{Z \geq Z_o \}$
	No $n < 30$ Distribución Normal	$t_o = \frac{(\overline{X-Y}) - D}{s_{(X-Y)}/\sqrt{n}}$	$H_0: \mu_{(X-Y)} = D$ vs $H_A: \mu_{(X-Y)} < D$	$(-t_{\alpha, n-1}, +\infty)$	$P\{t \leq t_o\}$
			$H_0: \mu_{(X-Y)} = D$ vs $H_A: \mu_{(X-Y)} > D$	$(-\infty, t_{\alpha, n-1})$	$P\{t \geq t_o\}$
			$H_0: \mu_{(X-Y)} = D$ vs $H_A: \mu_{(X-Y)} \neq D$	$(-t_{\alpha/2, n-1}, t_{\alpha/2, n-1})$	$P\{ t \geq t_o \} = 2P\{t \geq t_o \}$
$p_1 - p_2$ Muestras independientes	$n \geq 30$	$Z_o = \frac{\hat{p}_1 - \hat{p}_2 - D}{\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n} + \frac{\hat{p}_2(1-\hat{p}_2)}{m}}}$	$H_0: p_1 - p_2 = D$ vs $H_A: p_1 - p_2 < D$	$(-z_{\alpha}, +\infty)$	$P\{Z \leq Z_o\}$
			$H_0: p_1 - p_2 = D$ vs $H_A: p_1 - p_2 > D$	$(-\infty, z_{\alpha})$	$P\{Z \geq Z_o\}$
			$H_0: p_1 - p_2 = D$ vs $H_A: p_1 - p_2 \neq D$	$(-z_{\alpha/2}, z_{\alpha/2})$	$P\{ Z \geq Z_o \} = 2P\{Z \geq Z_o \}$

Siendo: $s_p = \sqrt{\frac{(n-1)s_X^2 + (m-1)s_Y^2}{n+m-2}}$

$$v = \frac{\left(\frac{s_X^2}{n} + \frac{s_Y^2}{m}\right)^2}{\frac{s_X^4}{n^2(n-1)} + \frac{s_Y^4}{m^2(m-1)}}$$

$$s_X^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

$$s_Y^2 = \frac{1}{m-1} \sum_{i=1}^m (y_i - \bar{y})^2$$